Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 1 / 方法 1 [NT02-M01]

Divisibility becomes simpler after a reversible change of variables. Reduce ratios, subtract multiples, and use the gcd to isolate the true scale or period; count only after these structural constraints are fixed.

整除在可逆换元后往往更简单：约分比例、减去倍数，并用最大公因数隔离真正尺度或周期；确定这些结构限制后再计数。

Use Euclidean reduction before listing divisors

先作欧几里得化简，再列约数

Procedure

1. Replace $A \mid B$ by $A \mid (B - qA)$ with a useful integer q.

2. For equal remainders, the modulus divides every pairwise difference. For $n + c \mid P(n)$, substitute $n = -c$ into the remainder.

3. List only divisors meeting the original positivity, digit or size restrictions.

4. Recover the original variables and check the remainder lies between zero and the modulus.

通用步骤

1. 以合适整数 q 将 $A \mid B$ 改写为 $A \mid (B - qA)$。

2. 余数相等时，模数整除所有两两差；对 $n + c \mid P(n)$，以 $n = -c$ 求余数。

3. 只列满足原正性、数位或大小限制的约数。

4. 还原原变量，并检查余数介于零与模数之间。

> Example. Find all positive integers n such that $n + 3$ divides $n^2 + 2n + 8$.

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 1 / 方法 1 [NT02-M01] (continued / 续)

> 例题：求全部满足 $n+3 \mid n^2+2n+8$ 的正整数 n。

Worked solution

1. Divide: $n^2+2n+8=(n+3)(n-1)+11$.

2. The condition is exactly $n+3 \mid 11$. Positivity requires $n+3 \geq 4$.

3. Only divisor 11 qualifies, giving $n=8$. Substitution gives 88, divisible by 11.

例题解析

1. 相除得 $n^2+2n+8=(n+3)(n-1)+11$。

2. 条件恰为 $n+3 \mid 11$，正性又要求 $n+3 \geq 4$。

3. 只有约数 11 合格，得 $n=8$；代入为 88，确可被 11 整除。

A remainder reduction is an equivalence; preserve both directions.

求余化简须保留充要关系。

Practice / 配套练习: AIME-2001-I-Q01 (WB p. 558), AIME-2006-I-Q03 (WB p. 564), AIME-2017-I-Q02 (WB p. 486), AIME-2020-I-Q04 (WB p. 488), AIME-2025-I-Q01 (WB p. 582), AIME-2025-II-Q02 (WB p. 490)

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 2 / 方法 2 [NT02-M02]

Reduce ratios and use coprimality to fix integer scale

约分比例并用互质性确定整数尺度

Procedure

1. Write every rational ratio in lowest terms before assigning an integer scale.

2. Use Euclid's lemma: if $\gcd(a,b)=1$ and $a \mid bc$, then $a \mid c$.

3. For equal fractions or a reduced numerator-denominator pair, resolve the whole common scale, not just one variable.

4. Check that the resulting counts or digits are feasible, since integrality alone may not suffice.

通用步骤

1. 设置整数尺度前，先把所有比例约至最简。

2. 使用欧几里得引理：若 $\gcd(a,b)=1$ 且 $a \mid bc$，则 $a \mid c$。

3. 分数相等或既约分子分母题，应确定整个公因尺度，不能只确定一个变量。

4. 检验所得人数或数位是否可实现，因为整数性未必充分。

> Example. A class has boys:girls $=4:7$. After five boys join, the ratio is $3:4$. Find the original class size.

> 例题：某班男女人数比为 $4:7$，加入五名男生后变为 $3:4$。求原班人数。

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 2 / 方法 2 [NT02-M02] (continued / 续)

Worked solution

1. Since $4, 7$ are coprime, write the original counts as $4t, 7t$ with positive integer t.

2. The new ratio gives $4(4t+5) = 3(7t)$, so $t = 4$.

3. The original size is $11t = 44$. The final counts $21, 28$ indeed have ratio $3:4$.

例题解析

1. 因 $4, 7$ 互质，原人数可写成 $4t, 7t$，其中 t 为正整数。

2. 新比例给出 $4(4t+5) = 3(7t)$，故 $t = 4$。

3. 原班共有 $11t = 44$ 人；最后人数 $21, 28$ 之比确为 $3:4$。

A reduced ratio makes the scale integral; an unreduced one may not.

最简整数比才保证对应尺度为整数。

Practice / 配套练习: AIME-2000-I-Q03 (WB p. 218), AIME-2012-I-Q06 (WB p. 73), AIME-2013-I-Q08 (WB p. 121), AIME-2015-II-Q01 (WB p. 485), AIME-2019-I-Q07 (WB p. 477), AIME-2022-II-Q01 (WB p. 489), AIME-2023-II-Q05 (WB p. 595)

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 3 / 方法 3 [NT02-M03]

Parametrize multiples and common-divisor constraints

参数化倍数与公约数限制

Procedure

1. If $a \mid b, c$, write $b = au, c = av$ before using a sum or product condition.

2. Combine repeated divisibility conditions with gcd or lcm. Distinguish a divisor bound from a multiple bound.

3. For a fixed parameter, count remaining positive solutions explicitly, or list bounded factor pairs.

4. Prove the parameter list complete and construct the extremal or counted objects.

通用步骤

1. 若 $a \mid b, c$，先写成 $b = au, c = av$，再代入和或积条件。

2. 用最大公因数或最小公倍数合并重复整除条件，区分约数界与倍数界。

3. 固定参数后，明确统计剩余正整数解，或列出有界因子对。

4. 证明参数列表完整，并构造相应最值对象或被统计对象。

> Example. Count positive ordered pairs (x, y) such that $x \mid y$ and $x + y = 24$.

> 例题：统计满足 $x \mid y$ 且 $x + y = 24$ 的正整数有序对 (x, y)。

Worked solution

1. Write $y = kx$ with $k \geq 1$. Then $x(k + 1) = 24$.

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 3 / 方法 3 [NT02-M03] (continued / 续)

2. Thus x can be any divisor of 24 no greater than 12: $1, 2, 3, 4, 6, 8, 12$.

3. Each determines the unique positive $y = 24 - x$ and satisfies divisibility. There are seven pairs.

例题解析

1. 写 $y = kx$，其中 $k \geq 1$，则 $x(k+1) = 24$。

2. 故 x 为 24 的不大于 12 的约数：$1, 2, 3, 4, 6, 8, 12$。

3. 每个 x 都确定唯一正整数 $y = 24 - x$ 并满足整除，共七对。

A parameter count must respect strict increase, positivity and ordering.

参数计数须同时尊重严格递增、正性与有序性。

Practice / 配套练习: AIME-2007-II-Q02 (WB p. 484), AIME-2007-II-Q07 (WB p. 164), AIME-2010-II-Q01 (WB p. 567), AIME-2010-II-Q06 (WB p. 44), AIME-2013-I-Q11 (WB p. 512), AIME-2014-I-Q06 (WB p. 45), AIME-2016-I-Q05 (WB p. 138), AIME-2016-II-Q04 (WB p. 659), AIME-2019-II-Q03 (WB p. 487), AIME-2026-II-Q01 (WB p. 141)

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 4 / 方法 4 [NT02-M04]

Use the gcd to find periods, orbit lengths and primitive segments

用最大公因数求周期、轨道长度与本原格段

Procedure

1. For motion modulo n by a fixed step a, the first return solves $n \mid ka$.

2. Dividing by $g = \gcd(a, n)$ gives orbit length n/g and g separate cycles.

3. For lattice displacement (u, v), divide by $\gcd(u, v)$ to get a primitive step; for reflections, compose them to get translations.

4. For unfolding or multiple modular coordinates, solve all return divisibilities simultaneously; correct shared endpoints when repeating a segment.

通用步骤

1. 模 n 每次走固定步长 a 时，首次返回满足 $n \mid ka$。

2. 约去 $g = \gcd(a, n)$，轨道长度为 n/g，共有 g 个循环。

3. 格点位移 (u, v) 除以 $\gcd(u, v)$ 得本原步长；反射则先复合成平移。

4. 展开反射或处理多个模坐标时，同时求解全部返回整除条件；重复格段须修正公共端点。

> Example. A token moves 8 positions each turn around 30 labeled positions. How many cycles does this rule form, and how long is each cycle?

> 例题：棋子在 30 个有标签位置组成的环上，每次前进 8 格。该规则形成多少个循环？每个多长？

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 4 / 方法 4 [NT02-M04] (continued / 续)

Worked solution

1. The return condition is $30 \mid 8k$. Since $\gcd(30, 8) = 2$, it is equivalent to $15 \mid k$.

2. Each cycle therefore contains 15 distinct positions before returning.

3. The 30 positions split into $30/15 = 2$ cycles, namely the two parity classes.

例题解析

1. 返回条件为 $30 \mid 8k$。因 $\gcd(30, 8) = 2$，等价于 $15 \mid k$。

2. 故每个循环返回前经过 15 个不同位置。

3. 30 个位置分为 $30/15 = 2$ 个循环，恰为奇偶两个类。

Multiplication of residue labels by a is a permutation iff gcd(a,n)=1. Addition by a is always a permutation and has one full orbit iff gcd(a,n)=1.

剩余类标签乘 a 为置换当且仅当 gcd(a,n)=1；加 a 总是置换，而只有 gcd(a,n)=1 时才形成单一满轨道。

Bézout gives $ua + vb = \gcd(a, b)$. Hence two integer periods a, b force their gcd to be a period. For reflected function values, retain the residual reflection on the resulting residue classes; period reduction alone does not count the final classes.

The segment from $(0, 0)$ to positive lattice point (m, n) crosses $m + n - \gcd(m, n)$ open unit-cell interiors. Start with one cell, add the $m - 1$ vertical and $n - 1$ horizontal grid crossings, and subtract the $\gcd(m, n) - 1$ interior lattice crossings counted in both directions. A simultaneous crossing enters only one new diagonal cell.

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 4 / 方法 4 [NT02-M04] (continued / 续)

裴蜀关系 $ua + vb = \gcd(a, b)$ 表明两个整数周期 a, b 会迫使其最大公因数也为周期。反射函数题在约化剩余类后仍须保留剩余反射，单有周期约化并未统计最终等价类。

从 $(0, 0)$ 到正整点 (m, n) 的线段穿过 $m + n - \gcd(m, n)$ 个单位格内部。先计首格，再加 $m - 1$ 次竖直与 $n - 1$ 次水平格线穿越，减去被两方向重复计算的 $\gcd(m, n) - 1$ 个内部整点穿越；同时穿越时只进入一个新的对角格。

Practice / 配套练习: AIME-2000-I-Q12 (WB p. 210), AIME-2002-I-Q11 (WB p. 386), AIME-2012-I-Q15 (WB p. 510), AIME-2016-I-Q14 (WB p. 403), AIME-2021-I-Q07 (WB p. 196), AIME-2023-I-Q14 (WB p. 705), AIME-2025-II-Q11 (WB p. 707), AIME-2007-II-Q05 (WB p. 692)

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 5 / 方法 5 [NT02-M05]

Track a gcd invariant through reduced-fraction updates

在既约分数更新中追踪最大公因数不变量

Procedure

1. Write the numerator and denominator update before reduction.

2. Find a linear combination preserved by that update, and express the reduction gcd through it and another changing coordinate.

3. After division by the gcd, update both auxiliary coordinates; list the exact indices at which cancellation occurs.

4. Prove no later return is possible when the invariant becomes a unit.

通用步骤

1. 先写约分之前的分子分母更新。

2. 寻找更新保持的线性组合，并用它与另一变化坐标表达约分公因数。

3. 除以公因数后，同时更新两个辅助坐标，并精确记录发生约分的下标。

4. 当不变量变成单位时，证明以后不会再次返回。

> Example. Reduce m/n after replacing it by $(m+2)/(n+3)$. If $d = n - m$ and $I = 3m - 2n$, express the next reduced d, I in terms of the cancellation gcd.

> 例题：将 m/n 更新为 $(m+2)/(n+3)$ 后约分。设 $d = n - m, I = 3m - 2n$，用约分公因数表达新 d, I。

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 5 / 方法 5 [NT02-M05] (continued / 续)

Worked solution

1. Before reduction the difference is $d+1$, while $3(m+2)-2(n+3)=I$.

2. The gcd is $g=\gcd(m+2,n+3)=\gcd(m+2,d+1)=\gcd(I,d+1)$, because $I=(m+2)-2(d+1)$.

3. Consequently $d_{new}=(d+1)/g$ and $I_{new}=I/g$. If $I=1$, every future gcd is one and d increases forever.

例题解析

1. 约分前差为 $d+1$，而 $3(m+2)-2(n+3)=I$ 保持不变。

2. 公因数为 $g=\gcd(m+2,n+3)=\gcd(m+2,d+1)=\gcd(I,d+1)$，因 $I=(m+2)-2(d+1)$。

3. 因此 $d_{新}=(d+1)/g, I_{新}=I/g$。若 $I=1$，以后公因数恒为一，而 d 永远递增。

Reduction changes the preserved linear combination by division; it is not literally constant across reduced states.

约分会把线性组合也除以公因数，它并非在既约状态间字面保持常数。

Practice / 配套练习: AIME-2021-I-Q10 (WB p. 491)

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 6 / 方法 6 [NT02-M06]

Combine divisor sets and divide a chain by its first term

组合约数集合，并用首项约化整除链

Procedure

1. The common divisors of A, B are exactly the divisors of $\gcd(A, B)$; an LCM target requires each full prime-power exponent to appear.

2. For subsets, count missing-exponent events and apply inclusion-exclusion.

3. For an increasing divisibility chain beginning at d, divide every later term by d to obtain a smaller-sum instance.

4. Keep base cases and empty/singleton objects explicit so a recurrence is exhaustive.

通用步骤

1. A, B 的公约数恰为 $\gcd(A, B)$ 的约数；达到某最小公倍数须让每个完整质数幂指数出现。

2. 对子集，统计缺少完整指数的事件并用容斥。

3. 对首项为 d 的递增整除链，将以后各项除以 d 得总和更小的问题。

4. 明确递推初值、空对象与单元素对象，保证穷尽。

> Example. Choose a subset of the six divisors of 12. How many subsets have LCM exactly 12?

> 例题：从 12 的六个正约数中选子集，有多少个子集的最小公倍数恰为 12?

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 6 / 方法 6 [NT02-M06] (continued / 续)

Worked solution

1. The divisors are $1, 2, 3, 4, 6, 12$. A successful subset needs a divisor carrying 2^2 and one carrying 3.

2. There are 2^6 subsets; 2^4 miss the full two-exponent, 2^3 miss the three-exponent, and 2^2 miss both.

3. Inclusion-exclusion gives $64 - 16 - 8 + 4 = 44$. The empty subset is automatically excluded by both missing conditions.

例题解析

1. 约数为 $1, 2, 3, 4, 6, 12$。合格子集需有携带 2^2 的约数及携带 3 的约数。

2. 全部为 2^6 个子集；缺完整二指数有 2^4 个，缺三指数有 2^3 个，两者均缺有 2^2 个。

3. 容斥得 $64 - 16 - 8 + 4 = 44$；空集已由两个缺失条件自动排除。

A recurrence requires a reversible first-step decomposition, not only a numerical pattern.

递推须来自可逆的首步分解，不能只凭数值规律。

For public-information candidates involving common multiples, first list all admissible bounded runs using the divisibility conditions. A person's uncertainty is a statement about how many surviving candidates contain their value; repeat the candidate elimination after each public announcement. The gcd/LCM arithmetic creates the initial list, while the elimination itself belongs to LO03.

对涉及公倍数的公开信息候选，先用整除条件列出全部有界连续区间。某人不能确定，表示含有其数值的存活候选不止一个；每次公开声明后须重新筛选候选。最大公因数与最小公倍数算术产生初始列表，筛选本身属于 LO03。

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 6 / 方法 6 [NT02-M06] (continued / 续)

Practice / 配套练习: AIME-2019-II-Q12 (WB p. 686), AIME-2025-II-Q07 (WB p. 492), AIME-2021-II-Q11 (WB p. 810)

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 7 / 方法 7 [NT02-M07]

Bound a factorial divisor before testing a quotient

检验阶乘商前先确定阶乘约数的界

Procedure

1. Factor the expression around one convenient factorial.

2. Use monotonicity of $n!$ to rule out all sufficiently large indices at once.

3. Test the few remaining indices by exact integer quotients.

通用步骤

1. 围绕方便的某个阶乘分解表达式。

2. 利用 $n!$ 的单调性一次排除所有足够大的下标。

3. 用精确整数商检验剩下的少量下标。

> Example. Find the largest n such that $n! \mid 10! + 9!$.

> 例题：求使 $n! \mid 10! + 9!$ 的最大 n。

Worked solution

1. Write $10! + 9! = 11 \cdot 9!$, so $9!$ divides the expression.

2. But division by $10!$ gives $11/10$, which is not an integer.

3. For $n \geq 11$, $n! \geq 11! = 110 \cdot 9! > 11 \cdot 9!$. Thus no larger index works, and the answer is 9.

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 7 / 方法 7 [NT02-M07] (continued / 续)

例题解析

1. 写成 $10! + 9! = 11 \cdot 9!$，故 $9!$ 可整除。

2. 除以 $10!$ 得 $11/10$，不是整数。

3. 当 $n \geq 11$，有 $n! \geq 11! = 110 \cdot 9! > 11 \cdot 9!$，所以更大下标均不可能，答案为 9。

Magnitude is a useful first filter; divisibility still needs an exact quotient at the boundary.

大小是有效的初筛，但边界处仍须检验精确整除。

Practice / 配套练习: AIME-2003-I-Q01 (WB p. 483)